# JUNIOR SCHOOL LEARNING HUB

GRADE 8 MATHEMATICS — SCHEME OF WORK — TERM 1 2027

School: _____ Teacher: _____ TSC No.: _____

CBC planning format | Lesson outcomes are organised into Knowledge, Skills and Attitudes. Learning experiences retain the activities in the supplied Grade 8 scheme.

| WK | LSN | STRAND | SUB-STRAND | LESSON LEARNING OUTCOMES | LEARNING EXPERIENCES | KEY INQUIRY QUESTIONS | LEARNING RESOURCES | ASSESSMENT METHODS | REFLECTION |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | Numbers | Integers - Identification of integers Integers - Representation of integers on number line | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define integers and distinguish them from non-integers<br>**Skills:** Identify positive integers, negative integers and zero in different situations<br>**Attitudes:** Appreciate the use of integers in daily life situations | **In groups, learners are guided to:**<br>• Discuss and find readings of thermometers showing positive and negative values<br>• Classify numbers as integers or non-integers<br>• Use real-life situations like floors above and below ground to represent integers | How do we identify integers in real life situations? | - Master Mathematics Grade 8, pg. 1 - Thermometers - Number cards - Charts with integers - Master Mathematics Grade 8, pg. 2 - Manila paper - Rulers - Markers - Number lines | - Observation - Oral questions - Written exercises | |
| 2 | 2 | Numbers | Integers - Addition of integers on number line Integers - Subtraction of integers on number line | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** State the rule for adding integers on a number line<br>**Skills:** Carry out addition of integers on a number line correctly<br>**Attitudes:** Value the importance of addition of integers in real life | **In groups, learners are guided to:**<br>• Use number cards and containers for selection<br>• Draw number lines on the ground<br>• Jump to the right to add positive numbers<br>• Mark and record positions after jumping | How do we carry out addition of integers? | - Master Mathematics Grade 8, pg. 3 - Number cards - Ground markings - Chalk - Counters - Master Mathematics Grade 8, pg. 4 - Number lines - Markers - Playground space | - Observation - Practical activities - Oral questions | |
| 2 | 3 | Numbers | Integers - Combined operations on number line Integers - Application of integers using IT resources | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe the order of combined operations on integers<br>**Skills:** Perform combined addition and subtraction of integers on number line<br>**Attitudes:** Show confidence in solving problems involving integers | **In groups, learners are guided to:**<br>• Practice mixed operations using number lines<br>• Solve problems involving temperature changes<br>• Work out problems involving floors in buildings | How do we perform combined operations of integers? | - Master Mathematics Grade 8, pg. 5 - Number lines - Temperature gauges - Real-life problem cards - Master Mathematics Grade 8, pg. 6 - Digital devices - Internet access - Integer games/apps | - Written exercises - Problem-solving tasks - Observation | |
| 2 | 4 | Numbers | Fractions - Order of operations in fractions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Identify the correct order of operations in fractions (BODMAS)<br>**Skills:** Carry out combined operations on fractions accurately<br>**Attitudes:** Appreciate the importance of following correct order | **In groups, learners are guided to:**<br>• Discuss and use correct order of operations in fractions<br>• Work through operations in brackets first, then multiplication/division, then addition/subtraction<br>• Copy operations on number cards and solve | How do we use fractions in real life situations? | - Master Mathematics Grade 8, pg. 8 - Fraction cards - Calculators - Charts showing BODMAS | - Written tests - Class activities - Oral questions | |
| 2 | 5 | Numbers | Fractions - Operations on fractions from shopping activities Fractions - Word problems involving fractions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain how fractions are used in shopping and trading<br>**Skills:** Work out fraction operations from shopping activities<br>**Attitudes:** Show responsibility in applying fractions to real situations | **In groups, learners are guided to:**<br>• Discuss and carry out operations on fractions from shopping and other real-life cases<br>• Role-play shopping scenarios<br>• Solve problems involving sharing and distribution | Where do we apply combined operations on fractions? | - Master Mathematics Grade 8, pg. 9 - Shopping lists - Price tags - Play money - Fraction pieces - Master Mathematics Grade 8, pg. 10 - Word problem cards - Fraction charts - Measuring tools | - Problem-solving - Practical activities - Written assignments | |
| 3 | 1 | Numbers | Fractions - Games and IT activities on fractions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe different games involving fractions<br>**Skills:** Use IT devices for learning operations on fractions and play games<br>**Attitudes:** Enjoy learning about fractions | **In groups, learners are guided to:**<br>• Play games of operations on fractions using IT devices or other resources<br>• Engage in drag and drop activities online<br>• Use fraction apps for practice | How can we make learning fractions more interesting? | - Master Mathematics Grade 8, pg. 11 - Tablets/computers - Internet access - Fraction games | - Observation - Game performance - Digital assessment | |
| 3 | 2 | Numbers | Fractions - Mixed practice on combined operations | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Recall the order of operations in fractions<br>**Skills:** Solve complex combined fraction operations proficiently<br>**Attitudes:** Show confidence in working with fractions | **In groups, learners are guided to:**<br>• Practice solving mixed fraction problems<br>• Work in groups on challenging fraction tasks<br>• Present solutions to the class | What strategies help us solve complex fraction problems? | - Master Mathematics Grade 8, pg. 12 - Exercise books - Fraction worksheets - Group work materials | - Written tests - Group presentations - Peer assessment | |
| 3 | 3 | Numbers | Fractions - Mixed practice on combined operations | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Recall the order of operations in fractions<br>**Skills:** Solve complex combined fraction operations proficiently<br>**Attitudes:** Show confidence in working with fractions | **In groups, learners are guided to:**<br>• Practice solving mixed fraction problems<br>• Work in groups on challenging fraction tasks<br>• Present solutions to the class | What strategies help us solve complex fraction problems? | - Master Mathematics Grade 8, pg. 12 - Exercise books - Fraction worksheets - Group work materials | - Written tests - Group presentations - Peer assessment | |
| 3 | 4 | Numbers | Fractions - Application and reflection | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Discuss various applications of fractions in daily life<br>**Skills:** Demonstrate mastery of fraction operations<br>**Attitudes:** Promote use of fractions in real life situations | **In groups, learners are guided to:**<br>• Discuss with peers, parents or guardians about areas where fractions are applied<br>• Share real-life experiences involving fractions<br>• Compile a portfolio of fraction work | How have fractions helped us in our daily lives? | - Master Mathematics Grade 8, pg. 13 - Portfolio materials - Reflection journals | - Portfolio assessment - Oral presentations - Self-assessment | |
| 3 | 5 | Numbers | Decimals - Conversion of fractions to decimals | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the relationship between fractions and decimals<br>**Skills:** Convert fractions to decimals using different methods<br>**Attitudes:** Appreciate the connection between fractions and decimals | **In groups, learners are guided to:**<br>• Practice converting fractions to decimals using equivalent fractions with denominators as powers of 10<br>• Use long division method to convert fractions<br>• Complete conversion tables | How do we convert fractions to decimals? | - Master Mathematics Grade 8, pg. 13 - Conversion charts - Calculators - Place value charts | - Written exercises - Oral questions - Class activities | |
| 4 | 1 | Numbers | Decimals - Identifying and converting recurring decimals | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define recurring and non-recurring decimals<br>**Skills:** Identify recurring decimals and convert them to fractions correctly<br>**Attitudes:** Show interest in working with recurring decimals | **In groups, learners are guided to:**<br>• Discuss and classify non-recurring and recurring decimals<br>• Indicate recurring digits using dot notation<br>• Practice converting recurring decimals to fractions using algebraic method | How do we identify and work with recurring decimals? | - Master Mathematics Grade 8, pg. 15 - Decimal cards - Number cards - Calculators | - Written tests - Practical exercises - Observation | |
| 4 | 2 | Numbers | Decimals - Rounding off decimals to decimal places | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** State the rules for rounding off decimals<br>**Skills:** Round off decimal numbers to required decimal places accurately<br>**Attitudes:** Value accuracy in rounding decimals | **In groups, learners are guided to:**<br>• Discuss and round off decimal numbers to required decimal places<br>• Practice rounding to 1, 2, 3 decimal places<br>• Use place value charts to understand rounding | How do we round off decimals correctly? | - Master Mathematics Grade 8, pg. 19 - Place value charts - Decimal number cards - Rounding worksheets | - Written assignments - Oral questions - Class tests | |

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| 4 | 3 | Numbers | Decimals - Expressing numbers in significant figures | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the meaning of significant figures<br>**Skills:** Write decimal and whole numbers to given significant figures<br>**Attitudes:** Show precision in expressing numbers | **In groups, learners are guided to:**<br>• Write decimal and whole numbers to given significant figures<br>• Discuss rules for identifying significant figures<br>• Practice expressing numbers to different significant figures | What are significant figures and why are they important? | - Master Mathematics Grade 8, pg. 21 - Number charts - Worksheets - Scientific calculators | - Written tests - Practical exercises - Observation | |
| 4 | 4 | Numbers | Decimals - Expressing numbers in standard form | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define standard form notation A × 10■<br>**Skills:** Write numbers in standard form correctly and convert them back<br>**Attitudes:** Appreciate the use of standard form for very large and small numbers | **In groups, learners are guided to:**<br>• Write numbers in standard form on learning materials such as cards or charts<br>• Practice expressing very large and very small numbers<br>• Understand the power of 10 notation | How do we express numbers in standard form? | - Master Mathematics Grade 8, pg. 23 - Standard form cards - Calculators - Charts | - Written exercises - Oral questions - Class activities | |
| 4 | 5 | Numbers | Decimals - Combined operations on decimals | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Identify the correct order of operations for decimals<br>**Skills:** Work out combined operations on decimals systematically<br>**Attitudes:** Show confidence in solving decimal problems | **In groups, learners are guided to:**<br>• Work out combined operations on decimals in the correct order<br>• Practice problems involving brackets, multiplication, division, addition and subtraction<br>• Solve complex decimal calculations | How do we perform combined operations on decimals? | - Master Mathematics Grade 8, pg. 24 - Operation cards - Calculators - Worksheets | - Written tests - Problem-solving - Observation | |
| 5 | 1 | Numbers | Decimals - Application of decimals to real life | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Identify situations where decimals are used in daily life<br>**Skills:** Apply decimals to solve practical problems<br>**Attitudes:** Promote use of decimals in daily activities | **In groups, learners are guided to:**<br>• Discuss and apply decimals to real life cases<br>• Solve problems involving money, measurements, temperature<br>• Work with real-life scenarios | Where do we use decimals in our daily lives? | - Master Mathematics Grade 8, pg. 26 - Real-life problem cards - Measuring instruments - Price lists | - Practical tasks - Written assignments - Oral presentations | |
| 5 | 2 | Numbers | Decimals - Application of decimals to real life | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Identify situations where decimals are used in daily life<br>**Skills:** Apply decimals to solve practical problems<br>**Attitudes:** Promote use of decimals in daily activities | **In groups, learners are guided to:**<br>• Discuss and apply decimals to real life cases<br>• Solve problems involving money, measurements, temperature<br>• Work with real-life scenarios | Where do we use decimals in our daily lives? | - Master Mathematics Grade 8, pg. 26 - Real-life problem cards - Measuring instruments - Price lists | - Practical tasks - Written assignments - Oral presentations | |
| 5 | 3 | Numbers | Decimals - Games and digital activities | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain how digital games enhance learning of decimals<br>**Skills:** Use IT devices to play games involving decimals<br>**Attitudes:** Enjoy learning about decimals through interactive activities | **In groups, learners are guided to:**<br>• Play games of operations on decimals using IT or other materials<br>• Use decimal apps and online games<br>• Engage in interactive decimal activities | How can technology enhance our understanding of decimals? | - Master Mathematics Grade 8, pg. 27 - Digital devices - Decimal games/apps - Internet access | - Observation - Game performance - Participation | |
| 5 | 4 | Numbers | Squares and Square Roots - Reading squares from tables | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain how to read mathematical tables for squares<br>**Skills:** Work out squares of numbers between 1.0 and 9.999 from tables<br>**Attitudes:** Show accuracy in using mathematical tables | **In groups, learners are guided to:**<br>• Read and write the squares of numbers from tables<br>• Practice locating numbers in the table and reading their squares<br>• Work through examples using Table 1.3 | What are squares of numbers? | - Master Mathematics Grade 8, pg. 29 - Mathematical tables - Number cards - Worksheets | - Practical exercises - Written tests - Observation | |
| 5 | 5 | Numbers | Squares and Square Roots - Squares of large numbers | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe the method for finding squares of numbers above 10<br>**Skills:** Work out squares of numbers above 10 using standard form and tables<br>**Attitudes:** Demonstrate systematic approach in calculations | **In groups, learners are guided to:**<br>• Practice finding squares of numbers above 10 using standard form method<br>• Convert numbers to standard form A × 10■<br>• Calculate squares and express in ordinary form | How do we find squares of numbers greater than 10? | - Master Mathematics Grade 8, pg. 33 - Mathematical tables - Standard form charts - Calculators | - Written exercises - Class activities - Oral questions | |
| 6 | 1 | Numbers | Squares and Square Roots - Squares of numbers less than 1 | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the process for squaring decimal numbers less than 1<br>**Skills:** Find squares of decimal numbers less than 1 using tables<br>**Attitudes:** Show precision in working with small numbers | **In groups, learners are guided to:**<br>• Practice finding squares of numbers less than 1<br>• Use standard form with negative powers of 10<br>• Apply systematic method for calculations | How do we find squares of numbers less than 1? | - Master Mathematics Grade 8, pg. 35 - Mathematical tables - Decimal cards - Worksheets | - Written tests - Practical exercises - Problem-solving | |
| 6 | 2 | Numbers | Squares and Square Roots - Reading square roots from tables | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain how to read square root tables<br>**Skills:** Work out square roots of numbers from 1 to 99.99 using tables<br>**Attitudes:** Appreciate the relationship between squares and square roots | **In groups, learners are guided to:**<br>• Read and write the square roots of numbers from tables<br>• Practice using Table 1.4 for square roots<br>• Add values from the ADD column correctly | Where do we apply square roots in real life? | - Master Mathematics Grade 8, pg. 37 - Mathematical tables - Square root charts - Number cards | - Written assignments - Oral questions - Class tests | |
| 6 | 3 | Numbers | Squares and Square Roots - Reading square roots from tables | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain how to read square root tables<br>**Skills:** Work out square roots of numbers from 1 to 99.99 using tables<br>**Attitudes:** Appreciate the relationship between squares and square roots | **In groups, learners are guided to:**<br>• Read and write the square roots of numbers from tables<br>• Practice using Table 1.4 for square roots<br>• Add values from the ADD column correctly | Where do we apply square roots in real life? | - Master Mathematics Grade 8, pg. 37 - Mathematical tables - Square root charts - Number cards | - Written assignments - Oral questions - Class tests | |
| 6 | 4 | Numbers | Squares and Square Roots - Square roots of large numbers | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe the method for finding square roots of numbers 100 and above<br>**Skills:** Find square roots of numbers 100 and above using tables<br>**Attitudes:** Show systematic approach in calculations | **In groups, learners are guided to:**<br>• Practice finding square roots of numbers above 100<br>• Use standard form method<br>• Work with both Table 1.4 and Table 1.5 appropriately | How do we find square roots of numbers above 100? | - Master Mathematics Grade 8, pg. 39 - Mathematical tables (Tables 1.4 & 1.5) - Worksheets - Calculators | - Written exercises - Practical work - Observation | |
| 6 | 5 | Numbers | Squares and Square Roots - Using calculators for squares and square roots | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Identify the square and square root functions on a calculator<br>**Skills:** Work out squares and square roots using a calculator correctly<br>**Attitudes:** Appreciate the efficiency of using calculators | **In groups, learners are guided to:**<br>• Practice working out squares and square roots using a calculator<br>• Compare calculator results with table results<br>• Use IT devices or other materials to play square and square root games | How do calculators help us find squares and square roots? | - Master Mathematics Grade 8, pg. 42 - Scientific calculators - Digital devices - Comparison worksheets | - Practical exercises - Observation - Written tests | |

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| 7 | 1 | Numbers | Rates, Ratio, Proportions and Percentages - Identifying rates | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define rate as a quotient relationship between two quantities<br>**Skills:** Identify rates in different real-life situations<br>**Attitudes:** Appreciate the use of rates in daily life | **In groups, learners are guided to:**<br>• Time while doing different activities such as calling using different mobile service providers<br>• Role play activities and note time taken<br>• Record and compare rates | How do we use rates in real life situations? | - Master Mathematics Grade 8, pg. 44 - Stopwatches - Rate cards - Mobile phones (for demonstration) | - Observation - Oral questions - Practical activities | |
| 7 | 2 | Numbers | Rates, Ratio, Proportions and Percentages - Working out rates | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the method for calculating rates<br>**Skills:** Calculate rates from given information accurately<br>**Attitudes:** Show precision in rate calculations | **In groups, learners are guided to:**<br>• Carry out activities to determine rates<br>• Calculate rates per unit time or quantity<br>• Solve rate problems from real-life contexts | How do we calculate rates from given information? | - Master Mathematics Grade 8, pg. 46 - Timers - Measuring tools - Rate worksheets | - Written tests - Problem-solving - Class activities | |
| 7 | 3 | Numbers | Rates, Ratio, Proportions and Percentages - Expressing fractions as ratios | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain how to convert fractions to ratios<br>**Skills:** Express fractions as ratios in simplest form<br>**Attitudes:** Value precision in ratio work | **In groups, learners are guided to:**<br>• Use cut outs from whole objects to relate fractions to ratios<br>• Practice writing fractions as numerator : denominator<br>• Simplify ratios to lowest terms | How do we express fractions as ratios? | - Master Mathematics Grade 8, pg. 48 - Cut-out materials - Ratio cards - Counters | - Written exercises - Practical work - Oral questions | |
| 7 | 4 | Numbers | Rates, Ratio, Proportions and Percentages - Comparing ratios | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe methods for comparing two or more ratios<br>**Skills:** Compare ratios using percentage method and LCM method<br>**Attitudes:** Show systematic approach in comparing ratios | **In groups, learners are guided to:**<br>• Discuss and compare ratios from cut outs<br>• Use LCM method to compare ratios<br>• Express ratios as percentages for easy comparison | How do we compare two or more ratios? | - Master Mathematics Grade 8, pg. 50 - Comparison charts - Ratio cards - Calculators | - Written tests - Class activities - Problem-solving | |
| 7 | 5 | Numbers | Rates, Ratio, Proportions and Percentages - Comparing ratios | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe methods for comparing two or more ratios<br>**Skills:** Compare ratios using percentage method and LCM method<br>**Attitudes:** Show systematic approach in comparing ratios | **In groups, learners are guided to:**<br>• Discuss and compare ratios from cut outs<br>• Use LCM method to compare ratios<br>• Express ratios as percentages for easy comparison | How do we compare two or more ratios? | - Master Mathematics Grade 8, pg. 50 - Comparison charts - Ratio cards - Calculators | - Written tests - Class activities - Problem-solving | |
| 8 | 1 | Numbers | Rates, Ratio, Proportions and Percentages - Division of quantities in ratios | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the process of dividing quantities in given ratios<br>**Skills:** Divide quantities in given ratios systematically<br>**Attitudes:** Show fairness in sharing quantities | **In groups, learners are guided to:**<br>• Discuss and share quantities of concrete objects in different ratios<br>• Use counters or bottle tops to practice sharing<br>• Solve sharing problems | How do we divide quantities using ratios? | - Master Mathematics Grade 8, pg. 51 - Counters - Bottle tops - Sharing materials | - Practical exercises - Written assignments - Observation | |
| 8 | 2 | Numbers | Rates, Ratio, Proportions and Percentages - Working out ratios | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Identify the method for finding ratios from given quantities<br>**Skills:** Work out ratios in different situations<br>**Attitudes:** Appreciate applications of ratios in daily life | **In groups, learners are guided to:**<br>• Calculate ratios from given quantities<br>• Find missing values in ratio problems<br>• Apply ratios to real situations | How do we determine ratios from given information? | - Master Mathematics Grade 8, pg. 53 - Data cards - Real-life examples - Worksheets | - Written tests - Problem-solving - Oral questions | |
| 8 | 3 | Numbers | Rates, Ratio, Proportions and Percentages - Increase and decrease using ratios | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain how ratios show increase or decrease in quantities<br>**Skills:** Work out increase and decrease of quantities using ratios<br>**Attitudes:** Apply ratio changes to real situations | **In groups, learners are guided to:**<br>• Discuss and determine increase and decrease using ratios<br>• Use the format new : old to express changes<br>• Solve problems involving ratio changes | How do ratios represent increase or decrease? | - Master Mathematics Grade 8, pg. 55 - Change scenario cards - Calculators - Worksheets | - Written exercises - Class activities - Problem-solving | |
| 8 | 4 | Numbers | Rates, Ratio, Proportions and Percentages - Percentage increase | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define percentage increase<br>**Skills:** Calculate percentage increase accurately using the formula<br>**Attitudes:** Show precision in percentage calculations | **In groups, learners are guided to:**<br>• Discuss and determine percentage increase of different quantities<br>• Use the formula: percentage change = (change/original) × 100%<br>• Solve real-life percentage problems | How do we calculate percentage increase? | - Master Mathematics Grade 8, pg. 57 - Percentage charts - Calculators - Problem cards | - Written tests - Practical exercises - Oral questions | |
| 8 | 5 | Numbers | Rates, Ratio, Proportions and Percentages - Percentage decrease | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define percentage decrease<br>**Skills:** Calculate percentage decrease correctly<br>**Attitudes:** Apply percentage decrease to real situations responsibly | **In groups, learners are guided to:**<br>• Work through percentage decrease problems<br>• Calculate new values after percentage decrease<br>• Solve problems involving discounts and reductions | How do we calculate percentage decrease? | - Master Mathematics Grade 8, pg. 58 - Discount cards - Price lists - Calculators | - Written assignments - Problem-solving - Class tests | |
| 9 | 1 | Numbers | Rates, Ratio, Proportions and Percentages - Identifying direct proportions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define direct proportion<br>**Skills:** Identify direct proportions in real life situations<br>**Attitudes:** Appreciate proportional relationships in daily activities | **In groups, learners are guided to:**<br>• Use IT devices or other materials to explore proportions<br>• Role play shopping activities to show direct relationships<br>• Identify situations where increase in one leads to increase in other | What is direct proportion? | - Master Mathematics Grade 8, pg. 59 - Proportion charts - Real-life examples - Digital devices | - Observation - Oral questions - Practical activities | |
| 9 | 2 | Numbers | Rates, Ratio, Proportions and Percentages - Working out direct proportions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the unitary method for solving direct proportion<br>**Skills:** Work out direct proportions systematically<br>**Attitudes:** Show accuracy in direct proportion calculations | **In groups, learners are guided to:**<br>• Complete tables showing direct proportional relationships<br>• Calculate missing values in direct proportion<br>• Apply direct proportion to solve problems | How do we solve direct proportion problems? | - Master Mathematics Grade 8, pg. 60 - Proportion tables - Worksheets - Calculators | - Written tests - Problem-solving - Class activities | |
| 9 | 3 | Numbers | Rates, Ratio, Proportions and Percentages - Working out direct proportions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the unitary method for solving direct proportion<br>**Skills:** Work out direct proportions systematically<br>**Attitudes:** Show accuracy in direct proportion calculations | **In groups, learners are guided to:**<br>• Complete tables showing direct proportional relationships<br>• Calculate missing values in direct proportion<br>• Apply direct proportion to solve problems | How do we solve direct proportion problems? | - Master Mathematics Grade 8, pg. 60 - Proportion tables - Worksheets - Calculators | - Written tests - Problem-solving - Class activities | |
| 9 | 4 | Numbers | Rates, Ratio, Proportions and Percentages - Identifying indirect proportions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define indirect proportion<br>**Skills:** Identify indirect proportions in different situations<br>**Attitudes:** Appreciate the difference between direct and indirect proportion | **In groups, learners are guided to:**<br>• Use hourglass to show and determine indirect relationships<br>• Identify situations where increase in one leads to decrease in other<br>• Practice with filling containers | What is indirect proportion? | - Master Mathematics Grade 8, pg. 62 - Hourglass - Containers - Bottle tops | - Observation - Practical work - Oral questions | |
| 9 | 5 | Numbers | Rates, Ratio, Proportions and Percentages - Working out indirect proportions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the method for solving indirect proportion<br>**Skills:** Work out indirect proportions systematically<br>**Attitudes:** Show understanding of inverse relationships | **In groups, learners are guided to:**<br>• Complete tables showing indirect proportional relationships<br>• Calculate values where ratios are inverted<br>• Solve time-speed-distance problems | How do we solve indirect proportion problems? | - Master Mathematics Grade 8, pg. 63 - Proportion worksheets - Calculators - Problem cards | - Written exercises - Problem-solving - Written tests | |

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| 10 | 1 | Numbers | Rates, Ratio, Proportions and Percentages - Application and reflection | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Discuss various applications of ratios and proportions<br>**Skills:** Apply ratios and proportions in various real-life contexts<br>**Attitudes:** Promote use of ratios and proportions in real life | **In groups, learners are guided to:**<br>• Watch videos on ratios and proportions as used in daily activities<br>• Discuss applications with parents or guardians<br>• Reflect on learning and compile portfolio | How do ratios and proportions help us in daily life? | - Master Mathematics Grade 8, pg. 64 - Video resources - Digital devices - Portfolio materials | - Portfolio assessment - Presentations - Self-assessment | |
| 10 | 2 | Algebra | Algebraic Expressions - Factorisation of algebraic expressions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define factorisation as the reverse of expansion<br>**Skills:** Identify the highest common factor (HCF) in algebraic expressions<br>**Attitudes:** Appreciate the use of factorisation in simplifying expressions | **In groups, learners are guided to:**<br>• Make three sets of cards showing algebraic expressions and their factored forms<br>• Match cards from different rows to form equations<br>• Discuss and identify common factors in terms<br>• Write HCF in front of brackets and remaining factors inside | How do we factorise algebraic expressions? | - Master Mathematics Grade 8, pg. 65 - Number cards - Algebraic expression cards - Charts | - Observation - Card matching activity - Oral questions | |
| 10 | 3 | Algebra | Algebraic Expressions - Identifying like and unlike terms in factorisation | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the concept of like and unlike terms<br>**Skills:** Find common factors for different sets of terms<br>**Attitudes:** Show systematic approach in identifying factors | **In groups, learners are guided to:**<br>• Discuss and identify like and unlike terms<br>• Find common factors from given sets of algebraic terms<br>• Practice factorising expressions with numerical and variable common factors<br>• Work in groups to factorise various expressions | What makes terms like or unlike in algebra? | - Master Mathematics Grade 8, pg. 67 - Factor cards - Worksheets - Group work materials | - Written exercises - Group presentations - Class activities | |
| 10 | 4 | Algebra | Algebraic Expressions - Simplification of algebraic fractions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the process of simplifying algebraic fractions<br>**Skills:** Simplify algebraic fractions by finding LCM of denominators<br>**Attitudes:** Value accuracy in simplifying fractions | **In groups, learners are guided to:**<br>• Discuss like and unlike terms in algebraic fractions<br>• Find LCM of denominators in algebraic fractions<br>• Combine fractions with different denominators<br>• Practice simplifying complex algebraic fractions | How do we simplify algebraic expressions? | - Master Mathematics Grade 8, pg. 68 - Fraction charts - LCM charts - Worksheets | - Written tests - Practical exercises - Problem-solving | |
| 10 | 5 | Algebra | Algebraic Expressions - Simplification of algebraic fractions | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the process of simplifying algebraic fractions<br>**Skills:** Simplify algebraic fractions by finding LCM of denominators<br>**Attitudes:** Value accuracy in simplifying fractions | **In groups, learners are guided to:**<br>• Discuss like and unlike terms in algebraic fractions<br>• Find LCM of denominators in algebraic fractions<br>• Combine fractions with different denominators<br>• Practice simplifying complex algebraic fractions | How do we simplify algebraic expressions? | - Master Mathematics Grade 8, pg. 68 - Fraction charts - LCM charts - Worksheets | - Written tests - Practical exercises - Problem-solving | |
| 11 | 1 | Algebra | Algebraic Expressions - Advanced simplification practice | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe steps for simplifying complex algebraic fractions<br>**Skills:** Simplify algebraic fractions involving multiple operations<br>**Attitudes:** Show confidence in working with algebraic fractions | **In groups, learners are guided to:**<br>• Practice writing fractions as single fractions<br>• Simplify fractions with algebraic denominators<br>• Solve problems involving algebraic fractions<br>• Work through real-life applications | What strategies help us simplify complex algebraic fractions? | - Master Mathematics Grade 8, pg. 69 - Practice worksheets - Real-life problem cards - Calculators | - Written assignments - Class tests - Oral questions | |
| 11 | 2 | Algebra | Algebraic Expressions - Using IT devices and application | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Identify IT resources for learning algebra<br>**Skills:** Use IT devices to work out algebra exercises and drag-drop activities<br>**Attitudes:** Enjoy using algebraic expressions in real life situations | **In groups, learners are guided to:**<br>• Use IT devices to work out exercises and activities in algebra<br>• Engage in drag and drop activities of grouping similar terms<br>• Play online games simplifying algebraic expressions<br>• Discuss applications with peers and parents | How can technology enhance our understanding of algebra? | - Master Mathematics Grade 8, pg. 71 - Digital devices - Internet access - Algebra apps/software | - Observation - Digital assessment - Participation | |
| 11 | 3 | Algebra | Linear Equations - Forming linear equations in two unknowns | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Define linear equations in two unknowns<br>**Skills:** Form linear equations from real-life situations using two variables<br>**Attitudes:** Show interest in forming equations from word problems | **In groups, learners are guided to:**<br>• Put masses on beam balance and add marbles to balance<br>• Give letters to represent unknowns<br>• Role play shopping activities to form equations<br>• Write equations from balancing scenarios | How do we solve linear equations in two unknowns? | - Master Mathematics Grade 8, pg. 72 - Beam balance - Masses (500g) - Marbles - Shopping scenario cards | - Observation - Practical activities - Oral questions | |
| 11 | 4 | Algebra | Linear Equations - More practice on forming equations | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Interpret word problems involving two unknowns<br>**Skills:** Form linear equations from various real-life scenarios<br>**Attitudes:** Appreciate the relevance of equations in daily life | **In groups, learners are guided to:**<br>• Write equations to represent ages, costs, and quantities<br>• Form equations from perimeter problems<br>• Create equations from problems involving animals and farming<br>• Practice with two-digit number problems | Where do we use linear equations in two unknowns in real life situations? | - Master Mathematics Grade 8, pg. 73 - Word problem cards - Real-life scenario cards - Worksheets | - Written exercises - Problem-solving - Class activities | |
| 11 | 5 | Algebra | Linear Equations - Solving by substitution method | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the substitution method for solving simultaneous equations<br>**Skills:** Solve linear equations in two unknowns using substitution systematically<br>**Attitudes:** Show precision in solving equations | **In groups, learners are guided to:**<br>• Write equations from fruit vendor scenario<br>• Name equations as (i) and (ii)<br>• Write one variable in terms of another<br>• Replace and simplify to find values of unknowns | How do we use substitution method to solve linear equations? | - Master Mathematics Grade 8, pg. 74 - Fruit pictures - Equation cards - Step-by-step charts | - Written tests - Practical exercises - Oral questions | |
| 12 | 1 | Algebra | Linear Equations - Advanced practice on substitution method | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe the complete process of substitution method<br>**Skills:** Solve complex simultaneous equations by substitution<br>**Attitudes:** Demonstrate mastery of substitution technique | **In groups, learners are guided to:**<br>• Practice solving equations with fractions using substitution<br>• Work through problems involving costs and quantities<br>• Solve problems about carpentry and furniture making<br>• Apply substitution to number problems | What are the key steps in substitution method? | - Master Mathematics Grade 8, pg. 75 - Practice worksheets - Real-life problem cards - Calculators | - Written assignments - Problem-solving - Class tests | |
| 12 | 2 | Algebra | Linear Equations - Advanced practice on substitution method | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Describe the complete process of substitution method<br>**Skills:** Solve complex simultaneous equations by substitution<br>**Attitudes:** Demonstrate mastery of substitution technique | **In groups, learners are guided to:**<br>• Practice solving equations with fractions using substitution<br>• Work through problems involving costs and quantities<br>• Solve problems about carpentry and furniture making<br>• Apply substitution to number problems | What are the key steps in substitution method? | - Master Mathematics Grade 8, pg. 75 - Practice worksheets - Real-life problem cards - Calculators | - Written assignments - Problem-solving - Class tests | |
| 12 | 3 | Algebra | Linear Equations - Solving by elimination method | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Explain the elimination method for solving simultaneous equations<br>**Skills:** Solve linear equations using elimination method systematically<br>**Attitudes:** Appreciate the efficiency of elimination method | **In groups, learners are guided to:**<br>• Form equations from shopping scenarios (plates and cups)<br>• Multiply equations to make coefficients equal<br>• Subtract corresponding parts to eliminate one variable<br>• Solve for remaining variable and substitute back | How do we solve equations using elimination method? | - Master Mathematics Grade 8, pg. 76 - Shopping scenario cards - Elimination charts - Step-by-step guides | - Written exercises - Practical work - Oral questions | |

| WK | LSN | STRAND | SUB-STRAND | LESSON LEARNING OUTCOMES | LEARNING EXPERIENCES | KEY INQUIRY QUESTIONS | LEARNING RESOURCES | ASSESSMENT METHODS | REFLECTION |
|---|---|---|---|---|---|---|---|---|---|
| 12 | 4 | Algebra | Linear Equations - More practice on elimination method | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Identify when to use elimination method<br>**Skills:** Solve various simultaneous equations by elimination efficiently<br>**Attitudes:** Show confidence in choosing appropriate methods | **In groups, learners are guided to:**<br>• Practice solving equations involving bread and tea leaves<br>• Work through problems with different coefficients<br>• Solve problems about costs of items<br>• Compare elimination and substitution methods | When is elimination method more suitable than substitution? | - Master Mathematics Grade 8, pg. 78 - Comparison charts - Practice worksheets - Method selection guides | - Written tests - Class activities - Problem-solving | |
| 12 | 5 | Algebra | Linear Equations - Application in real-life situations | **At the end of the lesson, the learner should be able to:**<br>**Knowledge:** Discuss various applications of linear equations in daily life<br>**Skills:** Apply linear equations to solve real-life problems involving rectangles, costs, and quantities<br>**Attitudes:** Recognize use of linear equations in real life | **In groups, learners are guided to:**<br>• Find sum and difference of two numbers using equations<br>• Solve problems about rectangular flower beds<br>• Work out problems involving hiring labourers<br>• Apply equations to school fees and shopping scenarios<br>• Watch videos on linear equations applications | How do linear equations help us solve real-life problems? | - Master Mathematics Grade 8, pg. 79 - Video resources - Real-life scenario cards - Digital devices - Application worksheets | - Portfolio assessment - Presentations - Written assignments - Self-assessment | |